# Renu Shekhawat

N-11, Department of Mathematics, Indian Institute of Science, Bengaluru, Karnataka - 560 012
renushekhawa@iisc.ac.in

## Employment

**Indian Institute of Science** — Bengaluru, India
Research Associate — Feb 2026 - Present
Department of Mathematics
Mentor : Prof. Tirthankar Bhattacharyya

## Education

**Indian Statistical Institute, Bangalore Centre** — Bengaluru, India
Philosophiae Doctorate in Mathematics — Jul 2019 - Feb 2026
Supervisor : Prof. Soumyashant Nayak

**University of Rajasthan** — Jaipur, India
Master of Mathematics — Aug 2016 - Aug 2018

**Sophia Girls' College** — Ajmer, India
Bachelor of Mathematics — Jul 2013 - Jul 2016

## Research Interests

Matrix analysis, Functional analysis, Operator algebras

## Publications / Preprints

- S. Nayak and R. Shekhawat; *On the Convergence of the Normalized Power Sequence of Spectral Operators*, J. Operator Theory 96, Issue 1, Summer 2026, 121–143. Available at : DOI:10.7900/jot.2025jan17.2511, https://arxiv.org/abs/2410.16318
- S. Nayak and R Shekhawat; *On the Jordan-Chevalley-Dunford decomposition of operators in type I Murray-von Neumann algebras*, available at : https://arxiv.org/abs/2506.17227

  (Submitted for publication).
- S. Nayak and R. Shekhawat; *On the Convergence of the Normalized Power Sequence of Riesz Operators*, available at : https://arxiv.org/abs/2606.13743

  (Submitted for publication).

## Thesis

*On the Jordan-Chevalley-Dunford Decomposition of Certain Classes of Operators and Convergence of Their Normalized Power Sequences.* (Submitted to Indian Statistical Institute in June 2025).

## Fellowships / Scholarships

- **Junior Research Fellowship**
  (Declined due to acceptance of the Institute fellowship at ISI)
  - University Grants Commission, Government of India. Dec 2018
  - Council of Scientific and Industrial Research, Department of Science and Technology, Government of India. Jun 2017
- **Inspire Scholarship for Higher Education**
  Department of Science and Technology, Government of India. Jul 2013 - Jul 2018

## Teaching Assistant

- Real Ananlysis, B.math(I) - Semester I Aug 2025 - Nov 2025
  Instructor : Prof. B. V. Rajarama Bhat
- Function spaces, B.math(III) - Semester I Aug 2023 - Nov 2023
  Instructor : Prof. Soumyashant Nayak
- Complex Analysis, M.Math(I) - Semester II Jan 2022 - May 2022
  Instructor : Prof. Ramesh Sreekantan
- Complex Analysis, B.Math(III) - Semester I Sep 2021 - Dec 2021
  Instructor : Prof. Mathew Joseph

## Conferences / Workshops Attended

- **41st Annual Conference of Ramanujan Mathematical Society (RMS) 2026** Bengaluru, India
  Manipal Institute of Technology, Bengaluru Jul 2 - 5, 2026
- **Young Mathematicians in Operator Algebras** Delhi, India
  Indian Statistical Institute, Delhi Centre Apr 13 – 18, 2026
- **Ergodic Group Actions and Unitary Representations III** Warsaw, Poland
  Institute of Mathematics of the Polish Academy of Sciences, Warsaw Feb 2 - 6, 2026
- School on **Invariant Structures in Group Actions** Będlewo, Poland
  Research and Conference Centre of the Institute of Mathematics of the Polish Academy of Sciences, Będlewo Jan 11 - 16, 2026
- **Young Mathematicians in $C^*$ Algebras (YM$C^*$A) 2025** Odense, Denmark
  University of Southern Denmark (SDU), Odense Jul 21 - 25, 2025
- **International Workshop on Operator Theory and its Applications (IWOTA) 2025** Enschede, Netherlands
  University of Twente, Enschede Jul 14 - 18, 2025

- **20th Workshop: Noncommutative Probability, Operator Algebras and Related Topics, with Applications** — Będlewo, Poland
  Research and Conference Centre of the Institute of Mathematics of the Polish Academy of Sciences, Będlewo — Jul 6 - 12, 2025
- **Young Mathematicians in Operator Algebras** — Delhi, India
  Indian Statistical Institute, Delhi Centre — Mar 24 – 28, 2025
- **Conference on Operator Algebra and Related Topics (COART)** — Mumbai, India
  Indian Institute of Technology, Bombay — Feb 24 – 29, 2025
- **Noncommutative Mathematics and Applications (NCMA)** — Bengaluru, India
  Indian Statistical Institute, Bangalore Centre — Oct 24 – 26, 2024
- **Young Mathematicians in Operator Algebras** — Delhi, India
  Indian Statistical Institute, Delhi Centre — Feb 26 – Mar 02, 2024
- **International Conference on Spectral and Approximation Theory (ICSAT)** — Kochi, India
  Cochin University of Science and Technology, Kerala — Nov 27 – 30, 2023
- **Conference on Functional Analysis and Related Topics (CFART)** — Mumbai, India
  Indian Institute of Technology, Bombay — Feb 21 – 25, 2023
- **IWM (Indian Women and Mathematics) Annual Conference** — Pune, India
  Indian Institute of Science Education and Research, Pune — Dec 27 - 29, 2022
- **Indo-French Centre for Applied Mathematics (IFCAM)**
  Summer School on Mathematical Aspects of Quantum Mechanics — Bengaluru, India
  Indian Institute of Science, Bengaluru — Jun 01 – 12, 2022

## Talks

- **Invited:**
  - For a short talk in *Functional Analysis Symposium* in **RMS-2026 41st Annual Conference of Ramanujan Mathematical Society**
    Manipal Institute of Technology, Bengaluru — Jul 4, 2026
  - For a minicourse, as an *early Career Researcher* in **Simons Semester "Invariant Structures in Groups, Spaces, and Algebras"**
    Institute of Mathematics of the Polish Academy of Sciences, Warsaw — Jan 11 - Feb 15, 2026
  - For a talk in **Analysis and Probability Research Group Seminar**
    Indian Institute of Science, Bengaluru — Sep 22, 2025
- **Contributed Talks :**
  - *A stronger form of Yamamoto's theorem on singular values*
    IWOTA 2025, University of Twente, Enschede — Jul 18, 2025

- *Jordan-Chevalley-Dunford decomposition in Type I Murray-von Neumann algebras*
  20th Workshop: Noncommutative Probability, Operator Algebras and Related Topics, with Applications, Będlewo — Jul 10, 2025
- *Convergence of the normalized power sequence for spectral operators*
  COART, Indian Institute of Technology, Bombay — Feb 27, 2025
- *On Matrices over $C(X)$ for $X$ a Stonean space*
  ICSAT, Cochin University of Science and Technology, Kerala — Nov 27, 2023

- **Seminar Talks :**
  - *An elementary computational approach to the Jordan-Chevalley decomposition and the Jordan canonical form.*
    PDF - RS Annual Symposium, Indian Statistical Institute, Bangalore Centre — Jan 31, 2025
  - *Convergence of the normalized power sequence for spectral operators*
    QP - Day at ISI, Indian Statistical Institute, Bangalore Centre — Jan 15, 2025